



CHINA'S INFLUENCE IN INDONESIA'S NICKEL SECTOR AND IMPLICATIONS FOR THE UNITED STATES

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The tensions between Indonesia, China, and the United States in the nickel market encapsulate the broader challenges shaping the geopolitics of critical minerals. Indonesia's resource nationalism, China's investment and technological lead in critical minerals processing, overcapacity, environmental issues, and the United States' weak industrial base all converge in the Indonesian nickel sector—making it a microcosm of the global competition over strategic materials. Indonesia's continued dominance of the global nickel industry, facilitated by Chinese investment and technological innovation, presents a complex strategic dilemma for the United States that Washington has been unable to successfully navigate to date. Ultimately, the United States faces a choice between difficult realities rather than ideal outcomes.

The Rise of Indonesia: From Stainless Steel to Battery Materials

Indonesia's rise in the global nickel market is inseparable from China's industrial strategy, investment capacity, and domestic demand. Jakarta's "downstreaming" policies combined with abundant nickel reserves created the framework for domestic processing. But Indonesia lacked the capital and technological capability required to build out an independent supply chain. China provided the inputs and turned Indonesia into the world's lowest-cost nickel producer while reshaping the economics of the entire industry.

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Two misconceptions pervade the story of Indonesia's rise as a nickel producer. First, despite headlines linking Indonesia's nickel boom to the rise of electric vehicles (EVs), the country's production began and remains oriented toward the stainless-steel sector. Globally, stainless steel accounts for approximately 70% of nickel demand, while EV batteries represent 10%–15%.¹ Indonesia's industry reflects this balance: most of its nickel output feeds China's stainless-steel supply chain rather than the battery market. Second, Indonesia's resource nationalism, particularly export bans on raw minerals, are sometimes viewed as the sole driver of its success in global supply chains for critical minerals. However, even before the 2014 export ban, Indonesia supplied roughly half of China's nickel ore imports, creating a level of dependency that made withdrawal costly for Chinese firms.² The throughline in Indonesia's rise as a nickel producer is China's presence in the industry providing both major investment and demand for nickel.³

China's investment in Indonesia followed Beijing's "going out" strategy—a state-backed push for Chinese firms to invest abroad. The strategy's goal was to secure supply chains, expand global influence, and support domestic economic growth by investing in natural resources, infrastructure, and strategic industries abroad. The strategy was supported by programs like the Belt and Road Initiative, while state-backed financial institutions offered lower-cost capital. The export ban and China's need for nickel to feed its stainless-steel sector, occurring simultaneously with a global surge in Chinese state financing, pushed Chinese firms to embed themselves within Indonesia's industrial policy.

China's initial investment in Indonesia's critical minerals sector focused on rotary kiln-electric furnace (RKEF) smelters producing nickel pig iron and ferronickel for stainless steel. Tsingshan, a Chinese metals and mining company specializing in stainless steel and nickel, became the anchor investor in Indonesia's new industrial era. In 2013, at the launch of the 21st-Century Maritime Silk Road, Tsingshan

signed a memorandum of understanding—witnessed by Xi Jinping and then Indonesian president Susilo Bambang Yudhoyono—to develop what became the Indonesia Morowali Industrial Park (IMIP). By 2015–17, the IMIP was operational, functioning not merely as a smelter complex but as a sprawling industrial city equipped with ports, coal-fired power plants, and dedicated logistics systems.

While investment was spearheaded by private firms, state support followed. Private-sector investment for the IMIP and other industrial parks was backed by Chinese state-directed financing, including the China-ASEAN Investment Cooperation Fund, Export-Import Bank of China, and Industrial and Commercial Bank of China. This coordinated, below-market financing ecosystem—unavailable to Western companies—enabled rapid capital deployment, high risk tolerance, willingness to accept environmental degradation, and construction timelines unmatched by any competitor. The IMIP is now the largest nickel industrial hub in the world, hosting at least fifteen operational smelters with a combined capacity of 500,000–700,000 tons of nickel per year, with another 300,000–500,000 tons planned.⁴ This combination of private and public investment successfully created a stable feedstock for China's domestic stainless-steel industry and drove global nickel prices downward.

Yet China's most consequential contribution came in the next phase: the commercialization of high-pressure acid leaching (HPAL) technology in Indonesia. It was long considered uneconomical to process the country's low-grade but abundant laterite deposits into higher-grade refined nickel material necessary for batteries. Western attempts in Australia and Papua New Guinea were plagued by cost overruns and engineering failures. However, Tsingshan (in partnership with GEM, CATL, and Hanwa) successfully started operations on two HPAL facilities in 2021. Huayou Cobalt and Lygend soon followed. Due to the Chinese firms' innovation and engineering know-how, the plants convert low-grade

laterite into mixed hydroxide precipitate (MHP), which is a key precursor material necessary for EV batteries. This unlocked a massive new supply stream that had previously been stranded, although with significant potential risks to surrounding ecosystems. Now over 80% of the country's battery-grade MHP output comes from Chinese-affiliated projects.⁵

In 2014, Indonesia had 3 smelters. More than ten years later, it has 49 RKEF smelters producing nickel pig iron and ferronickel for stainless steel and 5 operational HPAL facilities producing MHP for batteries with additional projects under construction.⁶ Nickel export revenues grew from \$11.9 billion in 2020 to more than \$38 billion by 2024, and now Indonesia accounts for roughly 60% of global nickel production.⁷ Looking ahead, the International Energy Agency projects that Indonesia's dominance will deepen, with the country expected to supply 67% of the world's mined nickel by 2030 and 75% by 2040.⁸ Its surge in supply has pushed global prices downward, forcing higher-cost competitors in countries like Australia and New Caledonia to put their facilities in care and maintenance, furthering consolidation of the market. Indonesia has also grown into the second-largest producer of cobalt, which is a byproduct of the HPAL process.⁹

Chinese firms control roughly 75% of Indonesia's refining capacity, with Tsingshan and Jiangsu Delong alone accounting for over 70%.¹⁰ Tsingshan underpins Indonesia's stainless-steel industry through the IMIP and is heavily involved in the battery supply chain through ownership in the Weda Bay Industrial Park, which is the second-largest nickel industrial park in Indonesia. Chinese influence is increasingly extending further downstream in Indonesia's battery supply chain through Huayou Cobalt, Lygend, and most notably CATL, the Chinese battery giant. CATL and partners broke ground in 2025 on an approximately \$6 billion integrated battery project that will span the entire battery supply chain from mining to battery recycling.¹¹

China's continued investment and innovation in Indonesia's critical minerals sector after trade restrictions were implemented should be a warning signal for Western policymakers. Though one school of thought believes that Chinese firms can be forced out of supply chains through trade restrictions or environmental standards, Indonesia's nickel industry shows Chinese firms' willingness to adapt to the political realities and find ways to make the system work in their favor. Additionally, policymakers should not underestimate the ability for Chinese firms to innovate across the supply chain. State financing or lower regulatory standards support them, but Chinese companies are also leaders in innovation.

Limited U.S. Engagement Entrenches China's Position in Indonesia

U.S. companies have a long history in copper and gold mining in Indonesia, but their engagement has diminished in response to resource nationalism. Freeport-McMoRan began operating the Grasberg copper-gold mine in the 1970s, which is one of the world's largest copper mines. Over time, however, Indonesia's growing emphasis on resource nationalism reshaped the operating environment. The Indonesian government forced the company to divest a majority stake in Grasberg in 2018 after years of domestic pressure. Freeport also committed to build a domestic copper smelter as a condition for retaining its mining rights. The other major U.S. actor in Indonesia, Newmont Mining, exited Indonesia entirely in 2016 after prolonged disputes over export restrictions, divestment mandates, and processing requirements. Mandatory divestment laws and overall uncertainty in the operating environment make it challenging for Western mining companies to enter the country at scale.

Indonesia's resource nationalism policies do not pose the same challenges for Chinese firms.

Whereas Western companies depend on stable regulations and predictable returns, China's hybrid ecosystem of state-owned and private enterprises can absorb volatility, accept low margins, and operate in environments that deter most foreign investors. Thus, Indonesia's policies have the effect of entrenching China's presence across the country's mining industry while also limiting the United States' role in a key Indo-Pacific country.

The only notable recent U.S. engagement in Indonesian nickel came from Ford, the all-American automaker. In 2023 the company announced a \$4.5 billion partnership with Vale Indonesia and China's Huayou Cobalt to build an HPAL facility in Indonesia. The project aims to produce 120,000 tons of MHP. Ford's move reflects the pressure facing Western automakers to secure critical minerals. But it is equally telling that Ford, despite geopolitical tensions, partnered with Huayou for the project to be successful.

Political pressures also limited U.S. engagement with Indonesia on critical minerals. The lack of a free trade agreement prevented Indonesian nickel from being eligible for the clean vehicle tax credit during the Biden administration, while pressure from senators and NGOs blocked a U.S.-Indonesia critical minerals agreement. Now, the oversized role of Chinese firms in Indonesia limits further engagement. China hawks claim that the political walls work to limit Chinese involvement in key supply chains, but these regulations further entrench the power of Chinese companies in Indonesia and prevent U.S. involvement in Indonesia's critical minerals sector.

The Unfortunate Reality

The unfortunate reality is that Indonesia's and the United States' priorities are aligned, but the United States' view of Chinese influence prevents further engagement. Indonesia wants to diversify its customer base. The Indonesian nickel market has been built largely by and for one customer,

China, but it no longer represents the largest growth market. Despite stainless steel being the largest sector of the nickel market, nickel products for EV batteries are the largest growth sector of the industry. But the Chinese EV market has pivoted to and is now dominated by cheaper, nickel-free, lithium-iron-phosphate (LFP) batteries.

Lithium-ion battery chemistries that include nickel in the cathode offer extended range but are more expensive than nickel-free chemistries. Despite the cost, the market in the United States remains focused on nickel-based batteries due to worries around range and the reliance on South Korean and Japanese battery producers that focus on nickel-based batteries. This means the United States, not China, is the largest growth market for nickel. Indonesia is keen to access this market, and regardless of the Trump administration's views on EVs, U.S. factories are already producing EV batteries that rely on nickel-based chemistries. For the United States, Indonesian nickel remains strategically important because it anchors global pricing and supply for a key input for stainless steel, defense-related alloys, and batteries.

Given the reluctance of U.S. mining companies to operate in Indonesia and the United States' push to expand domestic mineral processing, the most realistic path forward is not upstream investment but partnerships between Indonesian nickel producers and U.S. midstream firms or end users. Despite China's current influence in Indonesia, a growing number of mineral projects are not controlled by Chinese companies. Majority Chinese-owned battery-grade production is expected to decline from roughly 80% today to 54% by the early 2030s.¹² Additionally, the Indonesian government has demonstrated a willingness to mandate divestment. This creates an opportunity for U.S. engagement—particularly through offtake agreements, minority equity stakes, and joint ventures focused on precursor production or downstream conversion in the United States.

However, such engagement will not materialize without sustained government-to-government coordination to provide regulatory clarity, financing, and political cover for U.S. firms. These efforts could be embedded within broader trade or critical minerals agreements, allowing the United States to gain exposure to Indonesia's dominant nickel supply without abandoning its domestic industrial objectives. It is impossible to eliminate China's influence, but these steps could give the United States a foothold in a supply chain that it currently observes from the sidelines. Alternatively, the United States could decide China's influence over the nickel market is too great and U.S. companies should push forward on LFP batteries to limit the demand for nickel. However, there are still issues with this approach. The LFP supply chain is highly concentrated in China, which is responsible for nearly 99% of the global LFP cathode market, and Chinese firms CATL and BYD are leaders in LFP innovation. Whether the United States avoids Indonesian nickel and depends on a Chinese-dominated LFP supply chain or embraces Indonesian nickel and potentially depends on Chinese-affiliated HPAL, China retains decisive influence over pricing and supply.

This dilemma in U.S. policy toward Indonesia underscores the overarching challenges to securing supply chains of critical minerals and countering China's influence. As Western financing found Indonesia's resource nationalism too risky, state-backed Chinese investment paired with innovative companies led to overcapacity in a key supply chain that has left U.S. allies struggling and the United States forced to catch up, without a coherent strategy for achieving its goals.

U.S. Critical Minerals Strategy: What Does Diversification Mean?

The goal of efforts by the United States and other Western countries to secure supply chains of critical minerals and counter China's dominance

is diversification. The idea is to diversify supply chains away from China to ensure they cannot be weaponized, as Beijing has done through recent export restrictions on rare earth elements. The example of Indonesia's critical minerals industry raises fundamental questions regarding this strategy: Are supply chains considered "diversified" if production moves outside China, away from the influence of export restrictions? Or is diversification defined not by geography but by reduced dependence on Chinese corporate control altogether?

If diversification is only about geography, then Indonesia's export ban is beneficial for U.S. interests. If not for the export ban, the processing taking place in Indonesia would occur in China, which is the only other country with the midstream processing ecosystem to absorb and refine large volumes of nickel ore. The ore would not flow to the United States or its allies because the United States does not have the nickel-processing capability or the pre-cathode or cathode active materials production required for EV batteries. Even the only operational nickel mine in the United States exports ore to Canada for processing.

However, if diversification is broader than geography and includes measures to minimize Chinese influence, then the United States and its allies would need to build new supply chains excluding Chinese firms. One potential option to support such an endeavor is through a "green premium" to offset higher-cost but more sustainable production. Australian officials and producers have repeatedly called for differentiated markets, and the Australia-U.S. critical minerals framework included language on a "standards-based systems" critical minerals market.¹³ However, there is not strong market demand for a differentiated product, nor is there a widely accepted "sustainable" nickel standard. Buyers remain sensitive to costs rather than environmental, social, and governance attributes—particularly in an environment of chronic oversupply created by Indonesian expansion.

Building a separate non-Chinese nickel supply chain would be enormously costly and require a long-term industrial policy far more expansive than current U.S. efforts. Policymakers are therefore left with a difficult question: should scarce government financing be devoted to standing up a separate nickel supply chain or diverted toward other minerals and technologies where China's ability to weaponize the market is stronger?

Engaging directly with Indonesia through partnerships with nickel producers and increased

government coordination is the most efficient path forward and would allow policymakers to focus on immediate supply chain threats. Regardless of how policymakers ultimately resolve the question of prioritization, China's approach in Indonesia illustrates the importance of coordinated public-private investment, long-term capital, and industrial strategy in building resilient supply chains. These are lessons the United States will need to learn if it seeks meaningful diversification. ∞

Endnotes

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